



EXPERIENCE

Lead Researcher for Aalto Team

Aalto University & Nokia

📅 July 2024 – December 2024 📍 Espoo, Finland

- Analysis of Quantum Annealing & Ising machines with memristor-based hardware
- Development of model for Analog-Digital Ising Machine
- We efficiently solved multi-user MIMO problem with the model, which can provide further benefits on achieving 6G research goals

Ph.D. Researcher

Aalto University - Machine Learning for Health Group

📅 August 2019 – December 2024 📍 Espoo, Finland

- Development of coupled hidden Markov model (CHMM) to evaluate the dynamics of bacterial colonization and impact of treatments
- Development of mixture of CHMM to evaluate the difference in treatment responses of patients
- Modeling of blood glucose and insulin responses with multi-output hierarchical Gaussian Processes
- Development of sparse Kalman filtering for physical systems with nonlinear dynamics

M.Sc. Researcher

Boğaziçi University & BKM - Interbank Card Center

📅 June 2018 – June 2019 📍 İstanbul, Turkey

- Applied machine learning on "Big Data" structures
- Development of interpretable fraud detection algorithm
- Development of a tool to create fraud scores for workplaces

Data Scientist

Digital Helix - DeepC

📅 June 2018 – December 2018 📍 İstanbul, Turkey

- Supervised and unsupervised brain lesion detection on MRI scans
- Medical image segmentation, image classification and feature regression with deep learning models

M.Sc. Researcher

Boğaziçi University & Borusan - R&D

📅 February 2017 – April 2018 📍 İstanbul, Türkiye

- Unsupervised anomaly detection on wind turbines
- Development of several Bayesian models and deep learning models for anomaly detection in time-series structure

EDUCATION

🎓 Ph.D. in Computer Science

Aalto University

📅 August 2019 – December 2024

Thesis: Flexible Bayesian latent variable modeling of interacting processes in time-series healthcare

🎓 M.S. in Computational Science and Engineering

Boğaziçi University

📅 January 2017 – June 2019

Thesis: Anomaly Detection in Time Series

🎓 B.S. in Electrical and Electronics Engineering

Boğaziçi University

📅 September 2010 – June 2016

🎓 Degree in Science High School

İstanbul Atatürk Science High School

📅 September 2006 – June 2010

PROF. INTERESTS

- Statistical Models and Bayesian Inference
- Probabilistic Machine Learning
- Bayesian Deep Learning
- Reinforcement Learning
- Deep Generative Models
- Matrix and Tensor Factorizations

HONORS AND AWARDS

🚗 **Bualert** 📅 **2013 – 2017**
Alternative Energy Racing Team
– Design car electronics, ECU and energy management system

🏆 **Award** 📅 **2015**
Advanced Consultancy Training
– 2nd place on consultancy case study

🏆 **Award** 📅 **2011**
Global Management Challenge(GMC)
– Turkey 7th at national competition

🏆 **Award** 📅 **2010**
University Entrance Exam – Turkey
112th out of 2 million participant

PUBLICATIONS

Journal Articles

- Poyraz, Onur, Sini Heinonen, et al. (2024). "Computational modeling enables individual assessment of postprandial glucose and insulin responses after bariatric surgery". In: *medRxiv*, pp. 2024–11.
- Poyraz, Onur, Mohamad R. A. Sater, et al. (2022). "Modelling MRSA decolonization: interactions between body sites and the impact of site-specific clearance". In: *Journal of the Royal Society Interface* 19.191, p. 20210916. DOI: 10.1098/rsif.2021.0916.

Conference Proceedings

- Haitsiukevich, Katsiaryna et al. (2024). "Diffusion Models as Probabilistic Neural Operators for Recovering Unobserved States of Dynamical Systems". In: *2024 IEEE 34th International Workshop on Machine Learning for Signal Processing (MLSP)*, pp. 1–6. DOI: 10.1109/MLSP58920.2024.10734762.
- Poyraz, Onur and Pekka Marttinen (2023). "Mixture of Coupled HMMs for Robust Modeling of Multivariate Healthcare Time Series". In: *Proceedings of the 3rd Machine Learning for Health Symposium*. Vol. 225. PMLR, pp. 461–479.
- Poyraz, Onur, Cem Say, and Ali Taylan Cemgil (2020). "Merchant Risk Scoring". In: *Signal Processing and Communications Applications Conference (SIU), 2020 28th. IEEE*, pp. 1–4.
- Poyraz, Onur, Burak Kurutmaz, et al. (2018). "Anomaly Detection on Wind Turbines". In: *Signal Processing and Communications Applications Conference (SIU), 2018 26th. IEEE*, pp. 1–4.

VOLUNTEER WORK

Scientific Symposium, by Students for Students

May 2021 Espoo, Finland

Machine Learning at the Bosphorus İsmail Arı Summer School

July 2018 İstanbul, Turkey

COMPLETED COURSES

- Advanced Probabilistic Methods
- Bayesian Data Analysis
- Bayesian Statistics and Machine Learning
- Monte Carlo Methods
- Deep Learning
- Information Theory
- Kernel Methods
- Probabilistic Graphical Models
- Robot Learning
- Numerical Linear Algebra and It's Applications
- Computational Genomics
- Bioinformatics
- Complex Networks
- Computational Methods
- Interaction Design

LANGUAGES

Turkish (Native)
English
Finnish

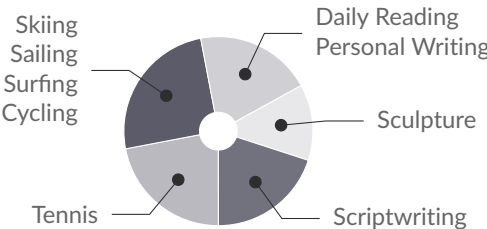


PROGRAMMING SKILLS

Python
PyTorch
Spark
SQL
R
Stan
Matlab
C/C++
Tensorflow
Julia



PERSONAL INTERESTS



REFERENCES

Area Product Owner & Program Manager

Antero Tossavainen

@ antero.tossavainen@nokia.com

Technology and Product Development
Nokia

Assoc. Professor Pekka Marttinen

@ pekka.marttinen@aalto.fi

Department of Computer Science
Aalto University

Professor Ali Taylan Cemgil

@ taylan.cemgil@boun.edu.tr

Research Scientist on DeepMind
Professor at Computer Engineering,
Bogazici University